ASVH-TRACE-001

Runtime Traceability Matrix

Mapping NHS England Guidance to Deterministic Runtime Authority Evaluation

Version: 1.0

Status: Engineering Working Draft

Project: ASVH-001 – NHS Ambient Scribing Runtime Verification Harness

1. Purpose

This document defines the end-to-end traceability model used by the **ASVH-001 NHS Ambient Scribing Runtime Verification Harness**.

Its purpose is to demonstrate how selected runtime-evaluable conditions contained within the NHS England *Guidance on the use of AI-enabled ambient scribing products in health and care settings* can be transformed into deterministic Runtime Authority evaluations immediately before AI-generated clinical documentation creates an institutional consequence.

The document does not reinterpret NHS guidance, create new governance requirements, or define clinical practice. Instead, it establishes a transparent engineering methodology that maps operational conditions described within the guidance to runtime context, deterministic evaluation rules, execution evidence and verification scenarios.

This document serves as the primary engineering reference for the ASVH project.

---

2. Scope

This document defines:

- the Runtime Traceability Methodology;
- the Runtime Context Catalogue;
- the Runtime Rule Catalogue;
- Authority Receipt traceability;
- Verification Scenario traceability;
- guidance-to-runtime mapping.

This document does **not**:

- replace NHS England guidance;
- interpret regulation;
- define clinical safety processes;
- determine clinical correctness;
- perform governance;
- specify procurement or deployment requirements.

---

3. Architectural Principle

The ASVH verification harness is based upon a single architectural principle:

> **Governance prepares the environment. Runtime Authority determines whether execution remains admissible immediately before institutional consequence formation.**

Accordingly, Runtime Authority evaluates only those conditions that exist immediately before an execution request is committed to a downstream institutional system.

---

4. Traceability Methodology

Every Runtime Authority rule shall satisfy the following traceability chain.

```None
NHS Guidance

↓

Runtime Context

↓

Runtime Rule

↓

Authority Receipt

↓

Verification Scenario
```

Each runtime rule shall be supported by one or more guidance references and shall produce deterministic evaluation outcomes.

---

5. Traceability Classification

Each runtime artefact is classified using one of three traceability categories.

5.1 Direct

A runtime condition explicitly described within the NHS guidance.

Example:

> Users review and approve AI-generated outputs before further actions.

5.2 Contextual

Operational information described within the guidance that forms part of the runtime evaluation context but is not itself a runtime decision.

Examples include:

- output type;
- consultation context;
- downstream destination;
- workflow identifiers.

5.3 Derived

An engineering control introduced to enable deterministic runtime evaluation while remaining consistent with the intent of the NHS guidance.

Derived controls do not extend the scope of the guidance and are clearly identified as implementation constructs.

---

6. Runtime Context Catalogue

Runtime Context represents observable operational facts available immediately before execution.

Runtime Context is descriptive rather than decision-making.

Context ID	Runtime Context	Type	Traceability
RC-001	Consultation Active	Boolean	Contextual
RC-002	Clinician Authenticated	Boolean	Contextual
RC-003	Output Reviewed	Boolean	Direct
RC-004	Output Approved	Boolean	Direct
RC-005	Intended Use	Enumeration	Direct
RC-006	Target Commit Path Available	Boolean	Direct
RC-007	Workflow Context	Enumeration	Contextual
RC-008	Execution Type	Enumeration	Derived
RC-009	Operational Context	Enumeration	Contextual
RC-010	Approved Product Status	Boolean	Contextual

These runtime context variables collectively define the operational state presented to Runtime Authority for evaluation.

---

7. Runtime Rule Catalogue

The Runtime Rule Catalogue identifies the deterministic Runtime Rules implemented by the ASVH Runtime Authority.

Each Runtime Rule evaluates a specific Runtime Context immediately before consequence formation to determine whether delegated authority remains legitimately exercisable.

The catalogue provides an engineering summary of each Runtime Rule, including its purpose, associated Runtime Context, expected evaluation outcomes, Authority Receipt evidence and verification scenario references.

The detailed normative behaviour of each Runtime Rule is defined separately within **ASVH-SPEC-001**.

The Runtime Rule Catalogue provides the authoritative engineering reference linking Runtime Context, Runtime Authority evaluation and execution evidence.

Rule ID	Rule Name	Primary Context	Outcome
RA-AS-001	Target Commit Path Available	RC-006	ALLOW / ESCALATE / REFUSE
RA-AS-002	Output Reviewed	RC-003	ALLOW / REFUSE
RA-AS-003	Output Approved	RC-004	ALLOW / REFUSE
RA-AS-004	Execution Type Identified	RC-008	ALLOW / ESCALATE / REFUSE
RA-AS-005	Intended Use Maintained	RC-005	ALLOW / ESCALATE / REFUSE
RA-AS-006	Operational Context Valid	RC-009	ALLOW / ESCALATE / REFUSE
RA-AS-007	Workflow Context Valid	RC-007	ALLOW / ESCALATE / REFUSE

7.1 RA-AS-001 — Target Commit Path Available

Purpose

RA-AS-001 verifies that the requested downstream execution path remains available immediately before consequence formation.

The rule prevents execution where the intended institutional destination cannot safely receive the requested consequence.

This evaluation ensures that Runtime Authority does not authorise execution toward an unavailable or invalid commit destination.

Primary Runtime Context

RC-006 — Target Commit Path Available

Traceability Classification

Direct

This Runtime Rule is derived directly from execution-related operational expectations identified within the NHS England guidance.

Runtime Authority Outcomes

Outcome	Description
ALLOW	Target commit path is available.
ESCALATE	Commit destination cannot be confirmed.
REFUSE	Commit destination is invalid or unavailable.

Authority Receipt Evidence

The Authority Receipt records:

- Commit Path Status
- Runtime Rule Identifier
- Evaluation Outcome
- Runtime Timestamp

Verification Scenarios

- AS-001
- AS-004

Engineering Notes

This Runtime Rule evaluates execution readiness rather than infrastructure availability.

7.2 RA-AS-002 — Output Reviewed

Purpose

RA-AS-002 verifies that AI-generated clinical documentation has undergone the required review before execution is permitted.

The Runtime Authority evaluates only the existence of the review state.

It does not assess the quality or correctness of the review.

Primary Runtime Context

RC-003 — Output Reviewed

Traceability Classification

Direct

Runtime Authority Outcomes

Outcome	Description
ALLOW	Output review completed.
REFUSE	Output review incomplete.

Authority Receipt Evidence

- Review Status
- Runtime Rule Identifier
- Evaluation Outcome

Verification Scenarios

- AS-001
- AS-002

Engineering Notes

Review completion is evaluated independently of clinical judgement.

7.3 RA-AS-003 — Output Approved

Purpose

RA-AS-003 confirms that the generated clinical documentation has received the required approval before commitment to the Electronic Patient Record.

Approval represents completion of the organisational workflow and forms part of the Runtime Context evaluated immediately before execution.

Primary Runtime Context

RC-004 — Output Approved

Traceability Classification

Direct

Runtime Authority Outcomes

Outcome	Description
ALLOW	Approval completed.
REFUSE	Approval not completed.

Authority Receipt Evidence

- Approval Status
- Runtime Rule Identifier
- Evaluation Outcome

Verification Scenarios

- AS-001
- AS-003

Engineering Notes

The Runtime Authority evaluates the presence of approval rather than the authority of the approver.

7.4 RA-AS-004 — Execution Type Identified

Purpose

RA-AS-004 verifies that the requested execution type is recognised and supported by the Runtime Authority.

Unsupported execution requests are prevented from progressing beyond the Execution Bind Point.

Primary Runtime Context

RC-008 — Execution Type

Traceability Classification

Derived

Runtime Authority Outcomes

Outcome	Description
ALLOW	Supported execution type.
ESCALATE	Execution type requires review.
REFUSE	Unsupported execution type.

Authority Receipt Evidence

- Execution Type
- Runtime Rule Identifier
- Evaluation Outcome

Verification Scenarios

- AS-001
- AS-007

Engineering Notes

Execution Type is an implementation construct introduced to support deterministic runtime evaluation.

7.5 RA-AS-005 — Intended Use Maintained

Purpose

RA-AS-005 verifies that the requested execution remains consistent with the intended operational use established for the generated output.

Execution beyond the approved operational purpose is prevented.

Primary Runtime Context

RC-005 — Intended Use

Traceability Classification

Direct

Runtime Authority Outcomes

Outcome	Description
ALLOW	Intended use maintained.
ESCALATE	Intended use cannot be confirmed.
REFUSE	Intended use exceeded.

Authority Receipt Evidence

- Intended Use Status
- Runtime Rule Identifier
- Evaluation Outcome

Verification Scenarios

- AS-001
- AS-005

Engineering Notes

This Runtime Rule evaluates execution purpose rather than clinical appropriateness.

7.6 RA-AS-006 — Operational Context Valid

Purpose

RA-AS-006 verifies that the operational environment remains valid immediately before execution.

Operational Context includes runtime conditions required to support deterministic execution.

Primary Runtime Context

RC-009 — Operational Context

Traceability Classification

Contextual

Runtime Authority Outcomes

Outcome	Description
ALLOW	Operational context valid.
ESCALATE	Operational context degraded.
REFUSE	Operational context invalid.

Authority Receipt Evidence

- Operational Context
- Runtime Rule Identifier
- Evaluation Outcome

Verification Scenarios

- AS-001
- AS-008

Engineering Notes

Operational Context represents runtime operating conditions rather than clinical workflow state.

7.7 RA-AS-007 — Workflow Context Valid

Purpose

RA-AS-007 verifies that the execution request remains consistent with the current workflow state immediately before consequence formation.

The Runtime Authority evaluates workflow continuity to ensure that execution occurs only within a valid operational sequence.

Primary Runtime Context

RC-007 — Workflow Context

Traceability Classification

Contextual

Runtime Authority Outcomes

Outcome	Description
ALLOW	Workflow state valid.
ESCALATE	Workflow state inconsistent.
REFUSE	Workflow state invalid.

Authority Receipt Evidence

- Workflow State
- Runtime Rule Identifier
- Evaluation Outcome

Verification Scenarios

- AS-001
- AS-006

Engineering Notes

Workflow Context validates execution sequencing rather than business process correctness.

7.8 Engineering Summary

The Runtime Rule Catalogue provides the authoritative engineering inventory of the deterministic Runtime Rules implemented by the ASVH Runtime Authority. Each Runtime Rule is linked to a defined Runtime Context, contributes structured evidence to the Authority Receipt and is exercised through one or more verification scenarios.

The catalogue establishes the engineering relationships between Runtime Context, Runtime Authority evaluation and runtime evidence while preserving a clear separation between engineering traceability and the normative Runtime Rule definitions provided in **ASVH-SPEC-001**.

---

8. Runtime Evaluation Model

8.1 Introduction

The Runtime Evaluation Model describes how the ASVH Runtime Authority applies the Runtime Rule Catalogue to determine whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

Whereas Chapter 7 identifies the individual Runtime Rules, this chapter describes the engineering process through which those rules are evaluated as a coherent runtime decision.

The Runtime Evaluation Model does not define implementation logic or prescribe software architecture. Instead, it establishes the deterministic evaluation sequence that every conforming implementation shall preserve.

The engineering objective is to ensure that identical Runtime Context consistently produces identical Runtime Authority outcomes, independent of implementation technology.

8.2 Engineering Objective

The Runtime Evaluation Model has four objectives:

- apply Runtime Rules consistently;
- preserve deterministic behaviour;
- maintain engineering traceability;
- produce reproducible Authority Receipt evidence.

Collectively, these objectives ensure that Runtime Authority decisions remain transparent, explainable and repeatable across different implementations.

8.3 Evaluation Sequence

Runtime Authority evaluation occurs immediately before execution reaches the institutional consequence.

The evaluation sequence is illustrated in Figure 8.1.

Figure 8.1 – Runtime Evaluation Sequence

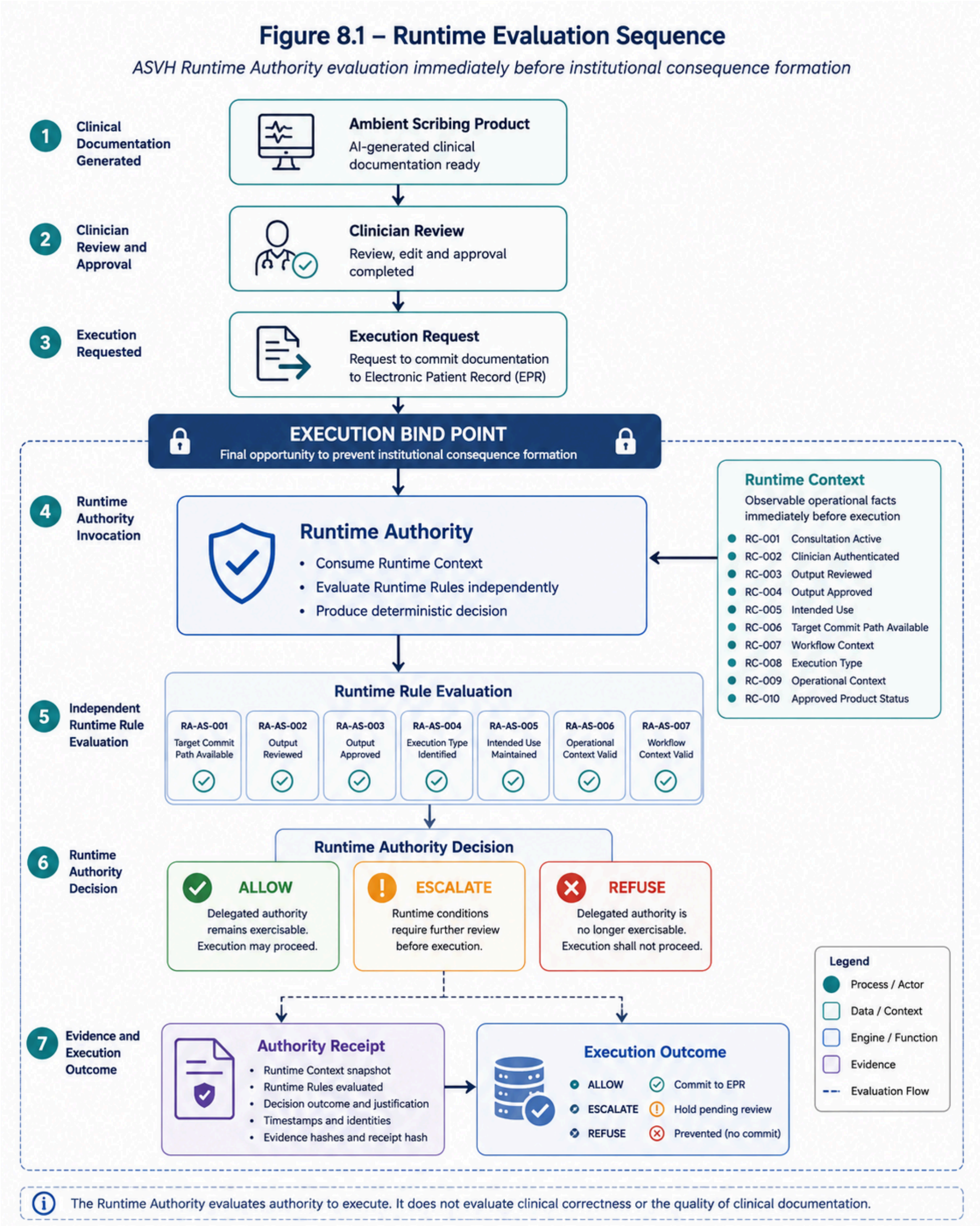

Figure 8.1 illustrates the Runtime Evaluation Model applied by the ASVH Runtime Authority immediately before institutional consequence formation. Following completion of clinician review and approval, an execution request reaches the **Execution Bind Point**, where execution is intentionally suspended. At this boundary, the Runtime Authority consumes the complete Runtime Context, evaluates the applicable Runtime Rules independently, and determines whether delegated authority remains legitimately exercisable. The resulting deterministic decision—**ALLOW**, **ESCALATE**, or **REFUSE**—is recorded within an Authority

Receipt before execution is either permitted to continue to the Electronic Patient Record or prevented from proceeding.

8.4 Runtime Context Consumption

The Runtime Authority consumes the Runtime Context described in Chapter 6.

Each Runtime Rule evaluates only the Runtime Context relevant to its engineering purpose.

The Runtime Authority does not modify Runtime Context during evaluation.

Instead, Runtime Context is treated as an immutable representation of the observable operational state immediately before execution.

This separation ensures that Runtime Authority decisions remain reproducible and independent of application behaviour.

8.5 Runtime Rule Evaluation

Each Runtime Rule is evaluated independently.

The Runtime Authority does not permit one Runtime Rule to modify the evaluation outcome of another.

Instead, every Runtime Rule contributes an individual engineering result that forms part of the overall Runtime Authority decision.

This design improves:

- explainability;
- traceability;
- verification;
- maintainability.

Independent Runtime Rule evaluation also simplifies future expansion of the Runtime Rule Catalogue without requiring changes to unrelated Runtime Rules.

8.6 Runtime Authority Decision

Following completion of Runtime Rule evaluation, the Runtime Authority determines the overall execution outcome.

The ASVH Runtime Evaluation Model supports three deterministic decisions.

Decision	Meaning
ALLOW	Delegated authority remains exercisable, and execution may proceed.
ESCALATE	Runtime conditions require further review before execution may continue.
REFUSE	Delegated authority is no longer exercisable, and execution shall not proceed.

These decisions are engineering outcomes derived from Runtime Rule evaluation rather than discretionary judgements made by the implementation.

8.7 Authority Receipt Generation

Completion of Runtime Authority evaluation results in generation of an Authority Receipt.

The Authority Receipt records:

- Runtime Context presented;
- Runtime Rules evaluated;
- Runtime Authority decision;
- execution timestamp;
- engineering traceability references;
- evidence metadata.

The Authority Receipt therefore provides the evidential record supporting the Runtime Authority decision and enables independent reconstruction of the evaluation.

8.8 Relationship to Verification

The Runtime Evaluation Model is exercised through the Verification Scenarios defined in Chapter 10.

Each Verification Scenario presents a defined Runtime Context to the Runtime Authority and confirms that deterministic Runtime Rule evaluation produces the expected Runtime Authority decision.

Verification therefore provides objective evidence that the Runtime Evaluation Model behaves consistently across representative operational conditions.

8.9 Relationship to Companion Publications

This chapter describes the engineering process used to evaluate Runtime Rules.

Companion publications provide additional detail for individual engineering artefacts.

- **ASVH-METH-001** describes the engineering methodology used to derive Runtime Rules.
- **ASVH-SPEC-001** defines the normative Runtime Rule specifications.
- **ASVH-ARCH-001** identifies the architectural location of Runtime Authority evaluation.
- **ASVH-001** demonstrates an executable implementation of the Runtime Evaluation Model.

8.10 Engineering Characteristics

The Runtime Evaluation Model exhibits five engineering characteristics.

Characteristic	Engineering Benefit
Deterministic	Identical Runtime Context produces identical Runtime Authority outcomes.
Independent	Runtime Rule evaluations remain isolated from one another.
Traceable	Every Runtime Rule contributes evidence to the Authority Receipt.
Implementation Independent	Evaluation behaviour is independent of implementation technology.

Reproducible	Runtime Authority decisions can be independently reconstructed.

8.11 Engineering Summary

The Runtime Evaluation Model defines how the Runtime Authority applies deterministic Runtime Rules to observable Runtime Context immediately before institutional consequence formation.

By evaluating Runtime Rules independently and recording the resulting evidence within the Authority Receipt, the model provides a transparent, reproducible and implementation-independent mechanism for determining whether delegated authority remains legitimately exercisable at the point of execution.

---

9. Authority Receipt Mapping

9.1 Introduction

Completion of Runtime Rule evaluation results in the generation of an Authority Receipt.

The Authority Receipt provides the permanent engineering record describing the Runtime Authority decision immediately before institutional consequence formation.

Unlike traditional application logging, the Authority Receipt is a structured engineering artefact designed to preserve the Runtime Context presented, the Runtime Rules evaluated, the resulting Runtime Authority decision and the evidence required to reconstruct that decision independently of the implementation.

The Authority Receipt therefore forms the evidential bridge between Runtime Authority evaluation and engineering traceability.

9.2 Engineering Purpose

The Authority Receipt serves five engineering purposes.

- preserve the Runtime Context presented for evaluation;
- record the Runtime Rules evaluated;
- capture the Runtime Authority decision;
- maintain engineering traceability;
- support deterministic reconstruction of the evaluation.

Collectively, these objectives ensure that Runtime Authority decisions remain explainable, reproducible and independently verifiable.

9.3 Authority Receipt Structure

Each Authority Receipt contains a consistent set of engineering information describing the runtime evaluation.

The engineering structure is illustrated in Figure 9.1.

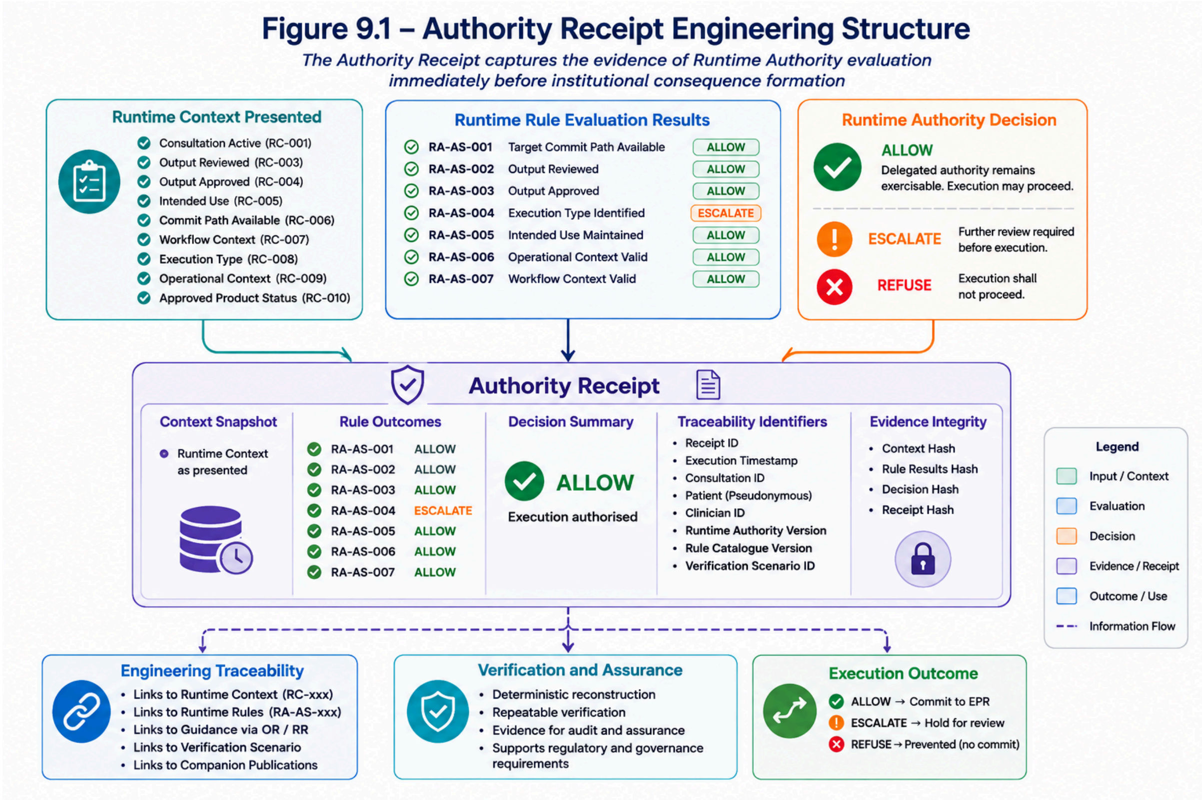

Figure 9.1 – Authority Receipt Engineering Structure

Figure 9.1 illustrates the engineering structure of the **Authority Receipt**, the permanent evidential artefact produced by the Runtime Authority immediately before institutional consequence formation. The diagram demonstrates how the Authority Receipt consolidates the complete outcome of runtime evaluation by combining the **Runtime Context** presented at the Execution Bind Point, the deterministic outcomes of each evaluated **Runtime Rule**, and the resulting **Runtime Authority Decision** into a single structured engineering record.

The Authority Receipt preserves five distinct categories of engineering evidence. First, it records an immutable snapshot of the Runtime Context exactly as presented to the Runtime Authority, ensuring that the operational state evaluated during execution can be reconstructed independently of the implementation. Second, it captures the individual outcomes of every Runtime Rule, providing transparent evidence of the deterministic evaluation process. Third, it records the resulting Runtime Authority decision—**ALLOW**, **ESCALATE**, or **REFUSE**—representing the authoritative determination of whether delegated authority remained legitimately exercisable immediately before execution.

In addition to the evaluation results, the Authority Receipt maintains a comprehensive set of engineering traceability identifiers linking the runtime evaluation to the wider ASVH publication series. These identifiers include Runtime Context references, Runtime Rule identifiers, verification scenario references, receipt identifiers, timestamps and implementation metadata, enabling every runtime decision to be traced back to the originating engineering artefacts.

The Authority Receipt also preserves evidence integrity through the inclusion of engineering metadata and cryptographic integrity values, ensuring that the recorded evidence can be verified and reconstructed without dependence on the executing application. Rather than functioning as a conventional application log, the Authority Receipt is designed as an implementation-independent engineering record that supports deterministic verification, audit, regulatory assurance and long-term evidence preservation.

Finally, the figure demonstrates that the Authority Receipt serves multiple engineering consumers. It provides the evidential foundation for engineering traceability, verification and assurance activities, while simultaneously documenting the execution outcome authorised by the Runtime Authority. By separating evidence generation from downstream execution, the Authority Receipt preserves the independence of the Runtime Authority and establishes a durable engineering record that can be independently reviewed long after execution has completed.

The Authority Receipt provides a single engineering record linking Runtime Context, Runtime Rule evaluation and Runtime Authority outcomes.

9.4 Runtime Context Mapping

The Authority Receipt records the Runtime Context presented immediately before evaluation.

Typical Runtime Context evidence includes:

Runtime Context	**Receipt Field**
Consultation Context	Consultation Active
Output Context	Output Reviewed
Output Context	Output Approved
Execution Context	Target Commit Path

Operational Context	Operational Context
Workflow Context	Workflow Context

The Runtime Authority records the Runtime Context exactly as presented during evaluation.

No Runtime Context values are modified during Authority Receipt generation.

9.5 Runtime Rule Mapping

The Authority Receipt records the outcome of every Runtime Rule evaluated during execution.

Each Runtime Rule contributes independent engineering evidence.

Runtime Rule	**Receipt Evidence**
RA-AS-001	Evaluation Outcome
RA-AS-002	Evaluation Outcome
RA-AS-003	Evaluation Outcome
RA-AS-004	Evaluation Outcome
RA-AS-005	Evaluation Outcome
RA-AS-006	Evaluation Outcome
RA-AS-007	Evaluation Outcome

Collectively, these Runtime Rule outcomes form the basis of the Runtime Authority decision.

9.6 Runtime Authority Decision Mapping

Following completion of Runtime Rule evaluation, the Runtime Authority records the resulting deterministic decision.

The Authority Receipt supports three Runtime Authority outcomes.

Decision	Receipt Representation
ALLOW	Execution authorised
ESCALATE	Further review required
REFUSE	Execution prevented

The Runtime Authority decision represents the aggregate engineering outcome of the Runtime Rule evaluation rather than an implementation-specific judgement.

9.7 Engineering Traceability

The Authority Receipt maintains direct engineering traceability by recording the identifiers associated with the runtime evaluation.

Typical traceability identifiers include:

- Runtime Rule Identifier
- Runtime Context Identifier
- Verification Scenario Identifier
- Authority Receipt Identifier
- Evaluation Timestamp

These identifiers enable every Runtime Authority decision to be traced back through the engineering artefacts described throughout the ASVH publication series.

9.8 Relationship to Verification

Every Verification Scenario produces an Authority Receipt.

Comparison of the generated Authority Receipt with the expected engineering evidence provides objective confirmation that Runtime Authority behaviour remains deterministic.

Verification therefore confirms both Runtime Rule behaviour and Authority Receipt completeness.

9.9 Relationship to Companion Publications

This chapter describes the engineering evidence represented within the Authority Receipt.

Companion publications provide additional detail.

- **ASVH-METH-001** describes the engineering methodology used to derive Runtime Context and Runtime Rules.
- **ASVH-TRACE-001** records the engineering relationships between Runtime Context, Runtime Rules and Authority Receipt evidence.
- **ASVH-SPEC-001** defines the Runtime Rules that contribute to the Authority Receipt.
- **ASVH-001** demonstrates an executable implementation of Authority Receipt generation.

9.10 Engineering Characteristics

The Authority Receipt exhibits five engineering characteristics.

Characteristic	Engineering Benefit
Deterministic	Identical Runtime Context produces identical Authority Receipts.
Traceable	Every Runtime Rule is represented within the receipt.
Independent	Receipt generation is independent of downstream execution.

Reproducible	Runtime evaluations can be reconstructed from recorded evidence.
Implementation Independent	Engineering evidence remains consistent across implementations.

9.11 Engineering Summary

The Authority Receipt provides the definitive engineering record of Runtime Authority evaluation immediately before institutional consequence formation.

By recording the Runtime Context presented, the Runtime Rules evaluated, the resulting Runtime Authority decision and the associated engineering traceability identifiers, the Authority Receipt enables deterministic reconstruction of runtime evaluations while preserving a clear separation between engineering evidence and implementation technology.

---

10. Verification Scenario Mapping

10.1 Introduction

Verification Scenarios provide the engineering mechanism through which the deterministic behaviour of the Runtime Authority is demonstrated and validated.

Each Verification Scenario presents a defined Runtime Context to the Runtime Authority and confirms that evaluation of the applicable Runtime Rules produces the expected Runtime Authority decision. The resulting Authority Receipt provides objective evidence that the runtime evaluation has been performed consistently and that the engineering relationships defined throughout this publication remain intact.

Verification Scenarios therefore provide the practical demonstration that the Runtime Evaluation Model behaves deterministically when presented with representative operational conditions.

10.2 Engineering Purpose

Verification Scenario Mapping serves five engineering objectives:

- demonstrate deterministic Runtime Authority behaviour;
- validate Runtime Rule implementation;
- confirm Authority Receipt generation;
- preserve engineering traceability;
- provide repeatable verification evidence.

Collectively, these objectives provide confidence that the Runtime Authority evaluates identical Runtime Context consistently across different executions and implementations.

10.3 Verification Model

Each Verification Scenario follows the same engineering sequence.

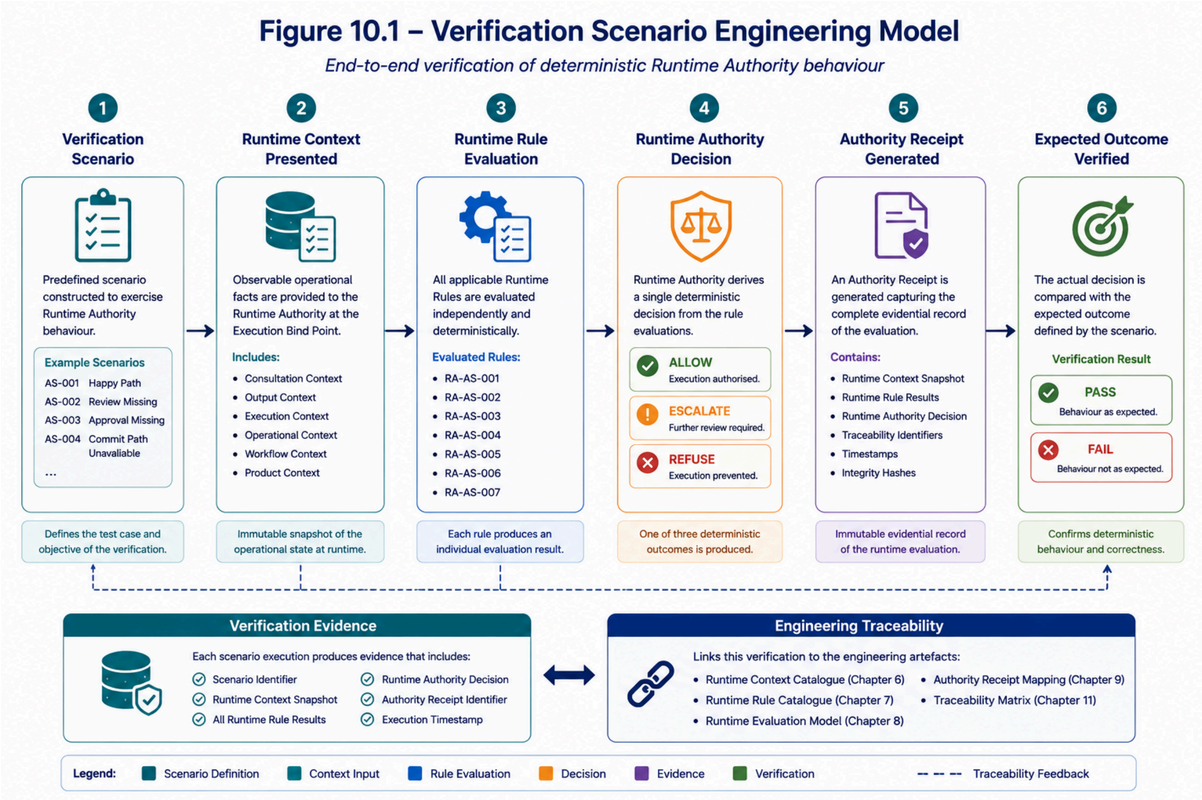

Figure 10.1 – Verification Scenario Engineering Model

Figure 10.1 illustrates the engineering lifecycle used to verify deterministic Runtime Authority behaviour within the ASVH Reference Implementation. Rather than validating individual software components in isolation, the Verification Scenario Engineering Model exercises the complete Runtime Authority evaluation process by presenting a predefined Runtime Context, executing the applicable Runtime Rules, generating a Runtime Authority decision, producing an Authority Receipt, and comparing the observed outcome with the expected engineering result.

The verification process begins with the definition of a **Verification Scenario**, which establishes a controlled operational condition designed to exercise one or more Runtime Rules. Each scenario specifies the Runtime Context presented to the Runtime Authority, the expected Runtime Authority decision, and the anticipated engineering evidence that should be produced. This provides a repeatable mechanism for validating deterministic behaviour under representative operational conditions.

The second stage presents the complete **Runtime Context** to the Runtime Authority at the Execution Bind Point. This context represents the observable operational state immediately before institutional consequence formation and includes the structured engineering domains defined throughout this publication, such as Consultation Context, Output Context, Execution Context, Operational Context and Workflow Context. The Runtime Context remains immutable throughout the evaluation process, ensuring that every verification exercise operates against a consistent engineering input.

During the third stage, the Runtime Authority evaluates all applicable **Runtime Rules** independently. Each Runtime Rule produces an individual engineering outcome which contributes to the overall Runtime Authority decision. Because every rule is evaluated deterministically against the presented Runtime Context, identical verification scenarios consistently produce identical evaluation results regardless of the implementation technology.

Following completion of Runtime Rule evaluation, the Runtime Authority determines one of three deterministic execution outcomes: **ALLOW**, **ESCALATE**, or **REFUSE**. This decision is immediately recorded within an **Authority Receipt**, together with the Runtime Context presented, the Runtime Rule outcomes, engineering traceability identifiers, timestamps and evidence metadata. The Authority Receipt therefore provides an immutable engineering record describing exactly how the Runtime Authority reached its decision.

The final stage compares the observed Runtime Authority decision with the expected outcome defined by the Verification Scenario. Successful agreement confirms that the Runtime Authority has behaved deterministically and that the engineering relationships between Runtime Context, Runtime Rules, Runtime Authority and Authority Receipt remain intact. Any deviation from the expected result indicates an engineering discrepancy requiring investigation rather than a discretionary operational judgement.

In addition to validating runtime behaviour, each Verification Scenario contributes structured **Verification Evidence** and reinforces the engineering traceability model established throughout the ASVH publication series. Every execution links directly to the corresponding Runtime Context, Runtime Rule Catalogue, Runtime Evaluation Model, Authority Receipt and Engineering Traceability Matrix, providing complete end-to-end reconstruction of the verification process.

10.4 Verification Scenario Catalogue

The ASVH Reference Implementation includes a series of deterministic Verification Scenarios designed to exercise representative runtime behaviour.

Scenario	Purpose	Expected Decision
AS-001	Happy Path	ALLOW
AS-002	Output Review Missing	REFUSE
AS-003	Output Approval Missing	REFUSE
AS-004	Commit Path Unavailable	ESCALATE
AS-005	Intended Use Exceeded	REFUSE

AS-006	Workflow Context Invalid	REFUSE
AS-007	Unsupported Execution Type	REFUSE
AS-008	Operational Context Invalid	REFUSE

The scenarios collectively exercise each Runtime Rule under representative operational conditions.

10.5 Runtime Rule Coverage

Each Verification Scenario exercises one or more Runtime Rules.

Runtime Rule	Verification Scenario(s)
RA-AS-001	AS-001, AS-004
RA-AS-002	AS-001, AS-002
RA-AS-003	AS-001, AS-003
RA-AS-004	AS-001, AS-007
RA-AS-005	AS-001, AS-005
RA-AS-006	AS-001, AS-008
RA-AS-007	AS-001, AS-006

This mapping confirms that every Runtime Rule is exercised through one or more Verification Scenarios.

10.6 Verification Evidence

Execution of each Verification Scenario produces a corresponding Authority Receipt.

Verification evidence includes:

- Runtime Context presented;
- Runtime Rules evaluated;
- Runtime Authority decision;
- Authority Receipt identifier;
- engineering traceability references;

- execution timestamp.

Verification evidence therefore demonstrates both deterministic behaviour and engineering completeness.

10.7 Relationship to Engineering Traceability

Verification Scenario Mapping completes the engineering traceability chain established throughout this publication.

Each Verification Scenario is linked directly to:

- Runtime Context;
- Runtime Rules;
- Runtime Authority decisions;
- Authority Receipt evidence.

This enables reviewers to reconstruct every verified Runtime Authority decision using the engineering artefacts defined within the ASVH publication series.

10.8 Relationship to Companion Publications

This chapter identifies the Verification Scenarios used to exercise the Runtime Authority.

Additional engineering detail is provided within the companion publications.

- **ASVH-METH-001** describes the methodology used to derive Runtime Requirements and Runtime Rules.
- **ASVH-TRACE-001** records the engineering relationships between Runtime Context, Runtime Rules and Verification Scenarios.
- **ASVH-SPEC-001** defines the normative Runtime Rule specifications.
- **ASVH-001** demonstrates the executable Verification Scenarios.

10.9 Engineering Characteristics

Verification Scenario Mapping exhibits five engineering characteristics.

Characteristic	Engineering Benefit
Deterministic	Identical Runtime Context produces identical verification outcomes.

Repeatable	Verification Scenarios may be executed consistently across implementations.
Traceable	Every verification result links directly to engineering artefacts.
Independent	Verification remains independent of implementation technology.
Evidence Based	Every scenario produces an Authority Receipt supporting objective review.

10.10 Engineering Summary

Verification Scenario Mapping provides the engineering mechanism through which the deterministic behaviour of the Runtime Authority is objectively demonstrated.

By presenting representative Runtime Context to the Runtime Authority, validating the resulting Runtime Rule evaluations and recording the generated Authority Receipts, Verification Scenarios establish repeatable engineering evidence supporting the Runtime Evaluation Model. Together with the Runtime Context Catalogue, Runtime Rule Catalogue and Authority Receipt Mapping, they complete the operational traceability required to reconstruct and verify Runtime Authority behaviour independently of the underlying implementation.

---

11. Engineering Traceability Matrix

11.1 Introduction

The Engineering Traceability Matrix provides the consolidated view of the engineering relationships established throughout the ASVH publication series.

Where the preceding chapters describe individual engineering artefacts, the Engineering Traceability Matrix records the complete lineage linking published NHS England guidance to deterministic Runtime Authority behaviour.

The matrix demonstrates that every Runtime Rule evaluated by the Runtime Authority can be traced to its originating engineering artefacts, including Operational Requirements, Runtime Requirements, Runtime Context, Verification Scenarios and Authority Receipt evidence.

The Engineering Traceability Matrix therefore represents the definitive engineering reference supporting reproducibility, auditability and independent verification.

11.2 Engineering Objective

The Engineering Traceability Matrix has five objectives:

- establish complete engineering lineage;
- demonstrate traceability between all engineering artefacts;
- support deterministic reconstruction of Runtime Authority decisions;
- provide evidence for engineering review and audit;
- preserve implementation independence.

Rather than describing implementation behaviour, the matrix documents the relationships between engineering artefacts produced during Runtime Requirement Engineering.

11.3 Engineering Traceability Model

The Engineering Traceability Model is illustrated in Figure 11.1.

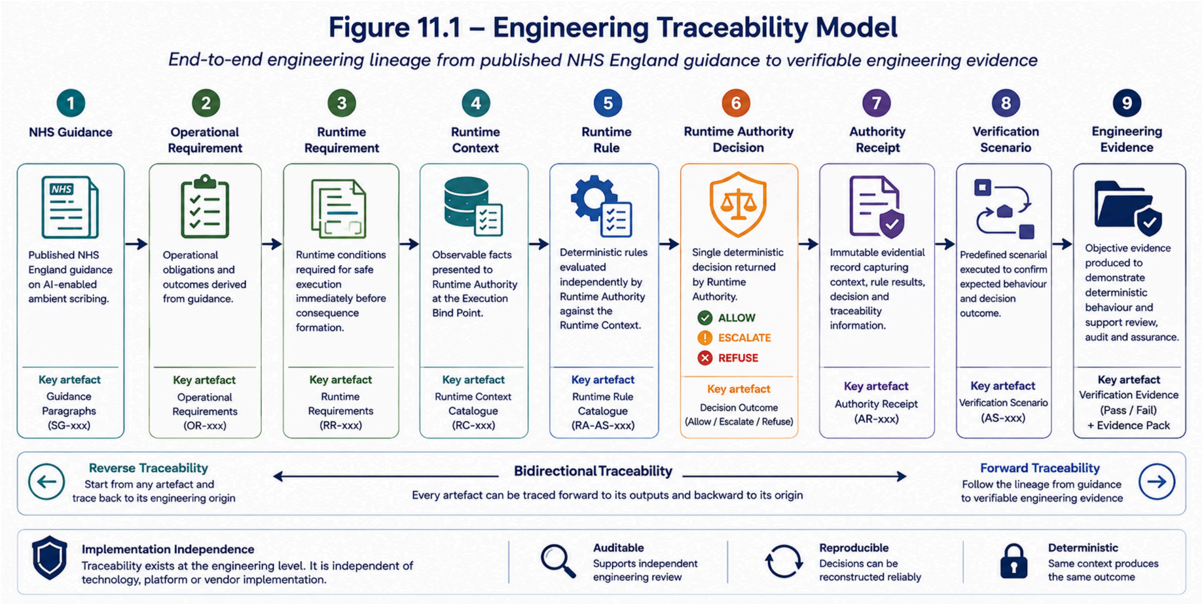

Figure 11.1 – Engineering Traceability Model

Figure 11.1 illustrates the complete engineering traceability model underpinning the ASVH Runtime Verification Framework. The model demonstrates the end-to-end engineering lineage through which published NHS England guidance is systematically transformed into deterministic runtime behaviour and ultimately into verifiable engineering evidence. Rather than representing a software workflow, the figure illustrates the relationships between the engineering artefacts created throughout the Runtime Requirement Engineering process and shows how those artefacts remain connected throughout the lifecycle.

The engineering process begins with the published **NHS England Guidance**, which provides the authoritative operational source material from which engineering obligations are identified. These obligations are formalised as **Operational Requirements**, representing the organisational behaviours and outcomes that the Runtime Authority must ultimately support. Operational Requirements are then refined into **Runtime Requirements**, which define the specific runtime conditions that must be evaluated immediately before institutional consequence formation.

The Runtime Requirements are subsequently represented as structured **Runtime Context**, providing the observable operational state presented to the Runtime Authority at the Execution Bind Point. Each Runtime Context element forms the input to one or more deterministic **Runtime Rules**, which are evaluated independently to determine whether delegated authority remains legitimately exercisable at the point of execution.

Completion of Runtime Rule evaluation results in a single **Runtime Authority Decision**, representing one of the deterministic outcomes defined by the ASVH Runtime Evaluation Model: **ALLOW**, **ESCALATE**, or **REFUSE**. This decision is immediately preserved within an immutable **Authority Receipt**, which records the Runtime Context presented, the Runtime Rule outcomes, the Runtime Authority decision and the associated engineering traceability identifiers required to reconstruct the evaluation independently of the implementation.

The Authority Receipt subsequently supports one or more **Verification Scenarios**, enabling the Runtime Authority's deterministic behaviour to be validated against predefined operational conditions. Successful execution of these scenarios produces objective **Engineering Evidence**, demonstrating that identical Runtime Context consistently produces identical Runtime Authority outcomes and that the engineering relationships defined throughout the publication series remain intact.

The lower section of the figure illustrates the bidirectional nature of engineering traceability. **Forward traceability** follows the engineering lineage from published guidance through to verifiable engineering evidence, demonstrating how operational intent is transformed into deterministic runtime behaviour. **Reverse traceability** enables any engineering artefact—including an Authority Receipt, Runtime Rule or Verification Scenario—to be traced back to its originating NHS England guidance. Together, these complementary views provide complete engineering lineage, supporting independent review, regulatory assurance, audit and deterministic reconstruction of Runtime Authority decisions.

The figure also highlights the fundamental engineering characteristics of the ASVH traceability model. Because traceability is maintained at the engineering level rather than within a specific software implementation, the relationships remain **implementation-independent**, allowing the methodology to be reproduced consistently across different technologies, platforms, and vendors. This provides an auditable, reproducible and deterministic foundation for Runtime Authority verification while preserving a clear separation between engineering methodology and implementation.

11.4 Engineering Traceability Matrix

The Engineering Traceability Matrix is the consolidated engineering index for the Runtime Verification Framework.

Each row represents a complete engineering lineage connecting published NHS England guidance to the Runtime Authority behaviour observed during execution.

Rather than documenting software implementation, the matrix records relationships between engineering artefacts.

Every Runtime Rule therefore has an identifiable origin, a deterministic runtime representation, and a verifiable evidence trail.

The Engineering Traceability Matrix consolidates the engineering relationships established throughout this publication.

NHS Guidance	Operational Requirement	Runtime Requirement	Runtime Context	Runtime Rule	Authority Receipt	Verification Scenario
SG-001	OR-001	RR-001	RC-006	RA-AS-001	Commit Path Status	AS-004
SG-002	OR-002	RR-002	RC-003	RA-AS-002	Review Status	AS-002
SG-003	OR-003	RR-003	RC-004	RA-AS-003	Approval Status	AS-003
SG-004	OR-004	RR-004	RC-008	RA-AS-004	Execution Type	AS-007
SG-005	OR-005	RR-005	RC-005	RA-AS-005	Intended Use	AS-005
SG-006	OR-006	RR-006	RC-009	RA-AS-006	Operation al Context	AS-008
SG-007	OR-007	RR-007	RC-007	RA-AS-007	Workflow Context	AS-006

The matrix provides a single engineering reference linking every Runtime Rule to its source guidance, engineering derivation and verification evidence.

11.5 Forward Traceability

Forward traceability demonstrates how published NHS England guidance progresses through the engineering lifecycle.

The sequence is:

- NHS England Guidance
- Operational Requirement
- Runtime Requirement
- Runtime Context
- Runtime Rule
- Runtime Authority Decision
- Authority Receipt
- Verification Scenario

Forward traceability confirms that every engineering artefact has a defined downstream purpose within the Runtime Verification process.

11.6 Reverse Traceability

Reverse traceability allows an engineering reviewer to begin with an Authority Receipt or Runtime Rule and reconstruct the engineering lineage back to the originating NHS England guidance.

This capability supports:

- engineering assurance;
- audit;
- independent review;
- deterministic reconstruction;
- evidence validation.

Reverse traceability is particularly valuable during post-execution analysis, where engineering evidence must demonstrate how a Runtime Authority decision was derived.

11.7 Relationship to Companion Publications

The Engineering Traceability Matrix connects all publications within the ASVH engineering series.

Publication	Engineering Responsibility
ASVH-METH-001	Runtime Requirement Engineering methodology
ASVH-TRACE-001	Engineering traceability relationships
ASVH-SPEC-001	Runtime Rule specification
ASVH-ARCH-001	Runtime Authority reference architecture
ASVH-IMPL-001	Reference implementation
ASVH-001	Executable Runtime Verification Harness

The Engineering Traceability Matrix therefore provides the navigational structure linking the complete publication series.

11.8 Engineering Characteristics

The Engineering Traceability Matrix exhibits six engineering characteristics.

Characteristic	Engineering Benefit
Complete	Every Runtime Rule is linked to its engineering origin.
Deterministic	Relationships remain consistent across executions.
Traceable	Engineering lineage can be reconstructed in both directions.
Independent	Traceability is independent of implementation technology.
Auditable	Engineering evidence supports external review.
Reproducible	Runtime Authority decisions can be reconstructed using documented engineering artefacts.

11.9 Engineering Summary

The Engineering Traceability Matrix provides the definitive engineering relationship model for the ASVH publication series.

By linking published NHS England guidance to Operational Requirements, Runtime Requirements, Runtime Context, Runtime Rules, Runtime Authority decisions, Authority Receipts and Verification Scenarios, the matrix establishes complete engineering lineage across the Runtime Verification process.

Together with the Runtime Context Catalogue, Runtime Rule Catalogue, Runtime Evaluation Model and Authority Receipt Mapping, the Engineering Traceability Matrix enables every Runtime Authority decision to be independently reconstructed, verified and audited while preserving implementation independence.

12. Design Constraints

12.1 Introduction

The Engineering Traceability Model described throughout this publication is founded upon a number of architectural and engineering constraints. These constraints preserve the deterministic behaviour of the Runtime Authority, maintain implementation independence, and ensure that engineering traceability remains consistent across different implementations.

The Design Constraints defined within this chapter represent engineering principles rather than implementation requirements. They establish the conditions necessary to preserve the integrity of Runtime Context, Runtime Rule evaluation, Authority Receipt generation and engineering traceability throughout the Runtime Verification process.

12.2 Deterministic Evaluation

Runtime Rule evaluation shall remain deterministic.

Identical Runtime Context shall always produce identical Runtime Rule outcomes and identical Runtime Authority decisions.

Implementation technology shall not influence evaluation behaviour.

This constraint ensures that Runtime Authority decisions remain reproducible and independently verifiable.

12.3 Immutable Runtime Context

The Runtime Context presented to the Runtime Authority shall be treated as immutable during evaluation.

Runtime Rule execution shall consume Runtime Context but shall not modify it.

This preserves the integrity of the Authority Receipt and enables independent reconstruction of the Runtime Authority decision.

12.4 Independent Runtime Rule Evaluation

Each Runtime Rule shall be evaluated independently.

The evaluation outcome of one Runtime Rule shall not modify or influence the evaluation logic of another Runtime Rule.

Independent Runtime Rule evaluation improves engineering transparency, simplifies verification and supports future expansion of the Runtime Rule Catalogue.

12.5 Separation from Implementation

Engineering artefacts defined within the ASVH publication series shall remain independent of implementation technology.

Runtime Context, Runtime Rules, Authority Receipts and engineering traceability shall preserve their meaning regardless of programming language, software architecture or deployment platform.

This separation enables multiple implementations to conform to the same engineering model.

12.6 Evidence Integrity

Authority Receipts shall accurately represent the Runtime Context presented, the Runtime Rules evaluated and the resulting Runtime Authority decision.

Authority Receipt generation shall not modify or reinterpret engineering evidence after Runtime Authority evaluation has completed.

This preserves the evidential integrity of the Runtime Verification process.

12.7 Complete Engineering Traceability

Every Runtime Rule shall maintain direct engineering traceability to:

- published NHS England guidance;
- Operational Requirements;
- Runtime Requirements;
- Runtime Context;
- Verification Scenarios;
- Authority Receipt evidence.

Engineering artefacts that cannot be traced through this chain shall not be considered part of the Runtime Verification Framework.

12.8 Implementation Independence

The engineering methodology described within the ASVH publication series shall remain implementation independent.

Alternative software implementations may differ in architecture, technology stack or deployment model, provided that they preserve:

- Runtime Context semantics;
- Runtime Rule behaviour;
- Runtime Authority decisions;
- Authority Receipt evidence;
- engineering traceability relationships.

Implementation independence ensures that conformance is measured against engineering behaviour rather than software design.

12.9 Relationship to Companion Publications

The Design Constraints described in this chapter establish engineering principles that apply consistently across the ASVH publication series.

- **ASVH-METH-001** defines the engineering methodology through which Runtime Requirements are derived.
- **ASVH-TRACE-001** records the engineering relationships between all runtime artefacts.
- **ASVH-SPEC-001** defines the normative Runtime Rule specifications.
- **ASVH-ARCH-001** describes the architectural placement of Runtime Authority.
- **ASVH-IMPL-001** demonstrates one implementation preserving these engineering constraints.

Collectively, these publications ensure that deterministic Runtime Verification can be reproduced consistently while maintaining architectural separation between methodology, specification, architecture and implementation.

12.10 Engineering Summary

The Design Constraints define the engineering principles necessary to preserve deterministic Runtime Verification within the ASVH Framework.

By enforcing deterministic evaluation, immutable Runtime Context, independent Runtime Rule execution, evidence integrity and implementation independence, these constraints ensure that Runtime Authority decisions remain reproducible, auditable and fully traceable regardless of implementation technology.

The Design Constraints therefore provide the engineering foundation that enables the Runtime Verification Framework to be applied consistently across different implementations while preserving the integrity of the engineering artefacts established throughout the ASVH publication series.

13. Conclusion

13.1 Engineering Summary

This publication has defined the engineering relationships that underpin the ASVH Runtime Verification Framework.

Beginning with published NHS England guidance, the document has demonstrated how Operational Requirements, Runtime Requirements, Runtime Context, Runtime Rules, Runtime Authority decisions, Authority Receipts and Verification Scenarios are connected through a single implementation-independent traceability model.

Rather than introducing additional governance controls or implementation-specific behaviour, the Engineering Traceability Model preserves the relationships between the engineering artefacts required to support deterministic Runtime Authority evaluation.

13.2 Engineering Contribution

The Engineering Traceability Model establishes complete engineering lineage across the Runtime Verification process.

Each Runtime Rule can be traced directly to the Runtime Context upon which it depends, the Operational and Runtime Requirements from which it was derived, the Verification Scenarios that exercise it and the Authority Receipt evidence generated during runtime evaluation.

This bidirectional traceability enables Runtime Authority decisions to be reconstructed independently of the implementation while preserving transparency, repeatability and engineering integrity.

13.3 Relationship to the ASVH Publication Series

ASVH-TRACE-001 forms the engineering relationship model within the ASVH publication series.

Together with the companion publications, it provides a complete engineering framework for Runtime Verification.

Publication	Engineering Responsibility

ASVH-METH-001	Defines the Runtime Requirement Engineering methodology used to derive engineering artefacts.
ASVH-TRACE-001	Records the engineering relationships between all Runtime Verification artefacts.
ASVH-SPEC-001	Defines the normative Runtime Rule specifications.
ASVH-ARCH-001	Describes the architectural placement of the Runtime Authority and Execution Bind Point.
ASVH-IMPL-001	Demonstrates one conforming implementation of the engineering framework.

Collectively, these publications separate engineering methodology, traceability, specification, architecture and implementation into distinct but complementary engineering references.

13.4 Implementation Independence

The Engineering Traceability Model is intentionally independent of implementation technology.

The engineering relationships described throughout this publication remain valid regardless of programming language, software platform, workflow engine or deployment architecture.

This separation ensures that conformance is measured against engineering behaviour rather than implementation design, allowing multiple implementations to preserve identical Runtime Verification semantics while maintaining architectural flexibility.

13.5 Final Observation

The ASVH Engineering Traceability Model demonstrates that deterministic Runtime Authority behaviour can be supported by a complete and auditable chain of engineering evidence extending from published operational guidance to runtime execution.

By preserving bidirectional traceability between guidance, requirements, Runtime Context, Runtime Rules, Runtime Authority decisions, Authority Receipts and Verification Scenarios, the model provides an implementation-independent engineering foundation for Runtime Verification.

This traceability enables every Runtime Authority decision to be independently understood, verified and reconstructed using documented engineering artefacts, supporting transparency, reproducibility and long-term engineering assurance across the ASVH Runtime Verification Framework.

Collectively, the ASVH publication series demonstrates how implementation guidance can be transformed into deterministic Runtime Authority behaviour while preserving complete engineering traceability from published guidance to runtime evidence.